Md Rezwan Bin Mizan

Experience

Research Assistant June 2024 – Present
Department of Mathematics, University of Houston

Graduate Teaching Assistant August 2021 – May 2024
Department of Mathematics, University of Houston

Lecturer January 2018 – July 2021
Department of Mathematics, American International University–Bangladesh (AIUB)

Education

PhD in Mathematics (In progress) Fall 2021 – Expected Graduation: Summer 2026
University of Houston, Texas, USA

- Advisor: Professor Maxim Olshanskii, Co-Advisor: Professor Ilya Timofeyev

MS in Applied Mathematics 2016 – 2017
University of Dhaka, Bangladesh

- GPA: 3.90/4.00
- Thesis: Spectral Relaxation Approach on Convective Heat Transfer of Magnetic and Non-Magnetic Particles over a Stretching Cylinder with Induced Magnetic Enhancement.

BS in Mathematics 2010 – 2015
University of Dhaka, Bangladesh

- CGPA: 3.70/4.00
- Project: A Study of Differential Equations in Population Biology.

HSC 2007 – 2009
Govt. Science College, Dhaka

- Major in Science

SSC 2007
Mirpur Siddhanto High School, Dhaka

- Major in Science

Research Interests

- Reduced Order Models (ROM)
- Computational Fluid Dynamics (CFD)
- Deep Learning and Time Series Analysis
- Numerical Optimization

Publications

A parametric tensor ROM for the shallow water dam break problem.
Authors: Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
***arXiv preprint**, arXiv:2506.20007, 2025.*

Parametric Reduced Order Models for the Generalized Kuramoto–Sivashinsky Equations.
Authors: Md Rezwan Bin Mizan, Maxim Olshanskii, and Ilya Timofeyev.
***arXiv preprint**, arXiv:2502.02718, 2025.*

Mathematical and Buongiorno Nanofluid Modelling to Predict the Thermal and Solutal Performance of Magneto-Bioconvective Casson Nanofluid Flowing in a Rotatory Disk.
Authors: Sulyman O. Salawu, MD Shamshuddin, Md Rezwan Bin Mizan, Sajjadul Bari.
*Accepted Manuscript, **Physica Scripta, IOP Publishing**, November 2, 2024. DOI: 10.1088/1402-4896/ad8e1a.*

Computation of Ferromagnetic/Nonmagnetic Nanofluid Flow Over a Stretching Cylinder with Induction and Curvature Effects.
Authors: Md Rezwan Bin Mizan, Mohammad Ferdows, MD Shamshuddin, O. Anwar Bég, Sulyman O. Salawu, Ali Kadir.
***Heat Transfer, Wiley**, March 13, 2021. DOI: 10.1002/htj.22122, 50(6), 5240–5266.*

Poster Presentations

Reduced Order Modeling of Kuramoto-Sivashinsky Equation using Proper Orthogonal Decomposition.
Mathematical Opportunities in Digital Twins (Math-DT), December 11-13, 2023, Van Meter Hall, George Mason University (Arlington Campus).
Presented the application of Proper Orthogonal Decomposition (POD) to build reduced-order models for the Kuramoto-Sivashinsky equation. Demonstrated the effectiveness of POD in capturing the essential dynamics of the system with reduced computational complexity.

POD–ROMs for the Generalized Kuramoto-Sivashinsky Equation (gKSE).
SIAM Conference Texas-Louisiana Section, 7th Annual Meeting, October 11-13, 2024, Hosted by Baylor University.
Presented a poster on Proper Orthogonal Decomposition (POD)-based Reduced Order Models (ROMs) for the gKSE, capturing spatiotemporal dynamics from chaotic to quasi-periodic regimes. This work tested single and multi-trajectory ROM approaches across a range of γ values, with metrics confirming the ROM's effectiveness in replicating complex dynamics.

Invited talks:

POD–ROMs for the Generalized Kuramoto-Sivashinsky Equation (gKSE).
Graduate Student Paper Presentation, October 11-13, 2024, Hosted by University of Houston.
Presented a poster on Proper Orthogonal Decomposition (POD)-based Reduced Order Models (ROMs) for the gKSE, capturing spatiotemporal dynamics from chaotic to quasi-periodic regimes.

This work tested single and multi-trajectory ROM approaches across a range of γ values, with metrics confirming the ROM's effectiveness in replicating complex dynamics.

Teaching Experience

Graduate Teaching Assistant August 2021 – Present
Department of Mathematics, University of Houston

- Offered tutoring assistance to undergraduate students in Calculus I, Calculus II, Calculus III, Discrete Mathematics, Linear Algebra and Probability.
- Helped students to grasp complex mathematical concepts, solve problems, and prepare for exams.

Lecturer January 2018 – July 2021
Department of Mathematics, American International University-Bangladesh (AIUB)

- Courses: Differential Calculus, Integral Calculus, Ordinary Differential Equations, Complex Variables, Numerical Methods, and Linear Algebra.

Leadership

Treasurer
AMS Chapter, University of Houston

- Managed financial records and transactions for the AMS UH Chapter.
- Assisted in organizing events and activities for the AMS UH Chapter.

Professional Training

TAMIDS Scientific Machine Learning (SciML) Summer School 2025
Organized by Texas A&M Institute of Data Science, Texas A&M University, College Station, TX, USA, May 12–16, 2025.

Self-Development Workshop on "Goal Setup, Presentation, Volunteerism, and First Step into Professional Development"
Organized by Life Carnival Foundation

Workshop on LaTeX: "Introduction to LaTeX"
Organized by Computer Science Engineering Department, University of Dhaka

Webinar on "Faculty Development Program on OBE"
Organized by American International University-Bangladesh (AIUB)

Skills

Programming Languages: Python, MATLAB, FORTRAN, Mathematica

Software Skills: Microsoft Office, LaTeX

Achievements

Graduate Tuition Fellowship 2021 – Present
Department of Mathematics, University of Houston

Teaching Assistantship 2021 – Present
Department of Mathematics, University of Houston

Bangladesh Sweden Trust Fund 2021
The Ministry of Finance, Bangladesh

National Science and Technology (NST) Fellowship 2016 – 2017
The Ministry of Science and Technology , Bangladesh